

Douglas C. Montgomery • George C. Runger

APPLIED STATISTICS AND PROBABILITY FOR ENGINEERS

7th Edition

WILEY

Applied Statistics and Probability for Engineers

Seventh Edition

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WILEY

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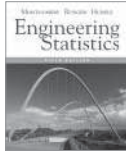
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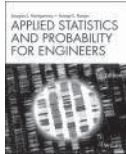
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by Montgomery and Runger

Introduction to engineering statistics, with topical coverage appropriate for either a one or two-semester course. An applied approach to solving real-world engineering problems.



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by Douglas C. Montgomery

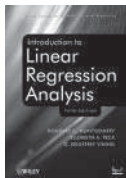
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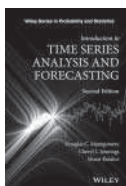


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by Myers, Montgomery and Vining

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An introductory text or reference on generalized linear models (GLMs). The range of theoretical topics and applications appeals both to students and practicing professionals.



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by Montgomery, Jennings and Kulahci

Methods for modeling and analyzing time series data, to draw inferences about the data and generate forecasts useful to the decision maker. Minitab and SAS are used to illustrate how the methods are implemented in practice. For Advanced undergraduate/first-year graduate, with a prerequisite of basic statistical methods. Portions of the book require calculus and matrix algebra.

Preface

INTENDED AUDIENCE

This is an introductory textbook for a first course in applied statistics and probability for undergraduate students in engineering and the physical or chemical sciences. These individuals play a significant role in designing and developing new products and manufacturing systems and processes, and they also improve existing systems. Statistical methods are an important tool in these activities because they provide the engineer with both descriptive and analytical methods for dealing with the variability in observed data. Although many of the methods we present are fundamental to statistical analysis in other disciplines, such as business and management, the life sciences, and the social sciences, we have elected to focus on an engineering-oriented audience. We believe that this approach will best serve students in engineering and the chemical/physical sciences and will allow them to concentrate on the many applications of statistics in these disciplines. We have worked hard to ensure that our examples and exercises are engineering- and science-based, and in almost all cases we have used examples of real data—either taken from a published source or based on our consulting experiences.

We believe that engineers in all disciplines should take at least one course in statistics. Unfortunately, because of other requirements, most engineers will only take one statistics course. This book can be used for a single course, although we have provided enough material for two courses in the hope that more students will see the important applications of statistics in their everyday work and elect a second course. We believe that this book will also serve as a useful reference.

We have retained the relatively modest mathematical level of the first five editions. We have found that engineering students who have completed one or two semesters of calculus and have some knowledge of matrix algebra should have no difficulty reading all of the text. It is our intent to give the reader an understanding of the methodology and how to apply it, not the mathematical theory. We have made many enhancements in this edition, including reorganizing and rewriting major portions of the book and adding a number of new exercises.

ORGANIZATION OF THE BOOK

Perhaps the most common criticism of engineering statistics texts is that they are too long. Both instructors and students complain that it is impossible to cover all of the topics in the book in one or even two terms. For authors, this is a serious issue because there is great variety in both the content and level of these courses, and the decisions about what material to delete without limiting the value of the text are not easy. Decisions about which topics to include in this edition were made based on a survey of instructors.

Chapter 1 is an introduction to the field of statistics and how engineers use statistical methodology as part of the engineering problem-solving process. This chapter also introduces the reader to some engineering applications of statistics, including building empirical models, designing engineering experiments, and monitoring manufacturing processes. These topics are discussed in more depth in subsequent chapters.

Chapters 2, 3, 4, and 5 cover the basic concepts of probability, discrete and continuous random variables, probability distributions, expected values, joint probability distributions, and independence. We have given a reasonably complete treatment of these topics but have avoided many of the mathematical or more theoretical details.

Chapter 6 begins the treatment of statistical methods with random sampling; data summary and description techniques, including stem-and-leaf plots, histograms, box plots, and probability plotting; and several types of time series plots. Chapter 7 discusses sampling distributions, the

central limit theorem, and point estimation of parameters. This chapter also introduces some of the important properties of estimators, the method of maximum likelihood, the method of moments, and Bayesian estimation.

Chapter 8 discusses interval estimation for a single sample. Topics included are confidence intervals for means, variances or standard deviations, proportions, prediction intervals, and tolerance intervals. Chapter 9 discusses hypothesis tests for a single sample. Chapter 10 presents tests and confidence intervals for two samples. This material has been extensively rewritten and reorganized. There is detailed information and examples of methods for determining appropriate sample sizes. We want the student to become familiar with how these techniques are used to solve real-world engineering problems and to get some understanding of the concepts behind them. We give a logical, heuristic development of the procedures rather than a formal, mathematical one. We have also included some material on nonparametric methods in these chapters.

Chapters 11 and 12 present simple and multiple linear regression including model adequacy checking and regression model diagnostics and an introduction to logistic regression. We use matrix algebra throughout the multiple regression material (Chapter 12) because it is the only easy way to understand the concepts presented. Scalar arithmetic presentations of multiple regression are awkward at best, and we have found that undergraduate engineers are exposed to enough matrix algebra to understand the presentation of this material.

Chapters 13 and 14 deal with single- and multifactor experiments, respectively. The notions of randomization, blocking, factorial designs, interactions, graphical data analysis, and fractional factorials are emphasized. Chapter 15 introduces statistical quality control, emphasizing the control chart and the fundamentals of statistical process control.

WHAT'S NEW IN THIS EDITION

It is highly gratifying that the sixth edition of the text was the most widely used edition in its history. For this seventh edition, we focused on revisions and enhancements aimed at *lowering costs to the students, better engaging them in the learning process, and providing increased support for instructors.*

Lower-Cost Product Formats

According to the publishing industry's most trusted market research firm, Outsell, 2015 represented a turning point in text usage: for the first time, student preference for digital learning materials was higher than for print, and the increase in preference for digital has been accelerating since.

While print remains important for many students as a pedagogical tool, the seventh edition is being delivered in forms that emphasize support for learning from digital materials. *All forms we are providing dramatically reduce the cost to students compared to the sixth edition.* These forms are:

- **WileyPLUS stand-alone.** The WileyPLUS course incorporates the full text and delivers a wealth of integrated, interactive learning aids. See the detailed list of the learning aids below.
- **Stand-alone e-text now with significant enhancements.** The e-text format for the seventh edition adds approximately 100 practice problems with solutions, several hundred basic "Check Your Understanding" exercises for student self-quizzing, and all of the videos included in the WileyPLUS course.
- The two electronic formats listed above can be bundled with an **Abridged Print Companion** that includes a loose-leaf copy of the entire text minus the exercises for a nominal additional cost over the any of the value-priced forms listed above.

To explore these options further or to discuss other options, contact your Wiley account manager (www.wiley.com/go/whosmyrep) or visit the product information page for this text on wiley.com. Instructors may also order bound print companions for the bundled package by contacting their Wiley account representative.

FEATURED IN THIS BOOK

Definitions, Key Concepts, and Equations

Throughout the text, definitions and key concepts and equations are highlighted by a box to emphasize their importance.

Learning Objectives

Learning Objectives at the start of each chapter guide the students in what they are expected to take away from this chapter and serve as a study reference.

LEARNING OBJECTIVES

After careful study of this chapter, you should be able to do the following:

- 1. Determine probabilities from probability density functions
- 2. Determine probabilities from cumulative distribution functions and cumulative distribution functions from probability density functions, and the reverse
- 3. Calculate means and variances for continuous random variables
- 4. Understand the assumptions for some common continuous probability distributions
- 5. Calculate probabilities, determine means and variances for some common continuous probability distributions
- 6. Approximate probabilities for binomial and Poisson distributions

Seven-Step Procedure for Hypothesis Testing

The text introduces a sequence of seven steps in applying hypothesis-testing methodology and explicitly exhibits this procedure in examples.

4.2 Cumulative Distribution Functions

An alternative method to describe the distribution of a discrete random variable can also be used for continuous random variables.

Cumulative Distribution Function

The **cumulative distribution function** of a continuous random variable X is

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(u) du \tag{4.3}$$

for $-\infty < x < \infty$.

The cumulative distribution function is defined for all real numbers.

EXAMPLE 4.3 | Electric Current

For the copper current measurement in Example 4.1, the cumulative distribution function of the random variable X consists of three expressions. If $x < 4.9$, $f(x) = 0$. Therefore,

$$F(x) = 0, \quad \text{for } x < 4.9$$

and

$$F(x) = \int_{-\infty}^x f(u) du = 5x - 24.5, \quad \text{for } 4.9 \leq x < 5.1$$

Finally,

$$F(x) = \int_{-\infty}^x f(u) du = 1, \quad \text{for } 5.1 \leq x$$

Therefore,

$$F(x) = \begin{cases} 0 & x < 4.9 \\ 5x - 24.5 & 4.9 \leq x < 5.1 \\ 1 & 5.1 \leq x \end{cases}$$

The plot of $F(x)$ is shown in Figure 4.6.

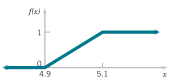


FIGURE 4.6
Cumulative distribution function for Example 4.3.

Notice that in the definition of $F(x)$, any $<$ can be changed to $\leq$ and vice versa. That is, in Example 4.3 $F(x)$ can be defined as either $5x - 24.5$ or 0 at the end-point $x = 4.9$, and $F(x)$ can be defined as either $5x - 24.5$ or 1 at the end-point $x = 5.1$. In other words, $F(x)$ is a continuous function. For a discrete random variable, $F(x)$ is not a continuous function. Sometimes a continuous random variable is defined as one that has a continuous cumulative distribution function.

The probability density function of a continuous random variable can be determined from the cumulative distribution function by differentiating. The fundamental theorem of calculus states that

$$\frac{d}{dx} \int_{-\infty}^x f(u) du = f(x)$$

Probability Density Function from the Cumulative Distribution Function

Given $F(x)$,

$$f(x) = \frac{dF(x)}{dx}$$

as long as the derivative exists.

9.1.6 General Procedure for Hypothesis Tests

This chapter develops hypothesis-testing procedures for many practical problems. Use of the following sequence of steps in applying hypothesis-testing methodology is recommended.

1. **Parameter of interest:** From the problem context, identify the parameter of interest.
2. **Null hypothesis, H_0 :** State the null hypothesis, H_0 .
3. **Alternative hypothesis, H_1 :** Specify an appropriate alternative hypothesis, H_1 .
4. **Test statistic:** Determine an appropriate test statistic.
5. **Reject H_0 if:** State the rejection criteria for the null hypothesis.
6. **Computations:** Compute any necessary sample quantities, substitute these into the equation for the test statistic, and compute that value.
7. **Draw conclusions:** Decide whether or not H_0 should be rejected and report that in the problem context.

Figures

Numerous figures throughout the text illustrate statistical concepts in multiple formats.

TABLE 11.1 Oxygen and Hydrocarbon Levels

Observation Number	Hydrocarbon Level $x(\%)$	Purity $y(\%)$
1	0.99	90.01
2	1.02	89.05
3	1.15	91.43
4	1.29	93.74
5	1.46	96.73
6	1.36	94.45
7	0.87	87.59
8	1.23	91.77
9	1.55	99.42
10	1.40	93.65
11	1.19	93.54
12	1.15	92.52

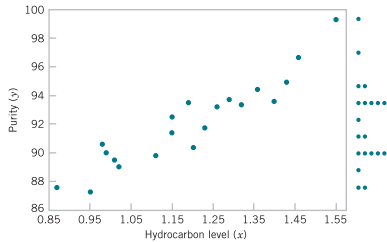


FIGURE 11.1 Scatter diagram of oxygen purity versus hydrocarbon level from Table 11.1.

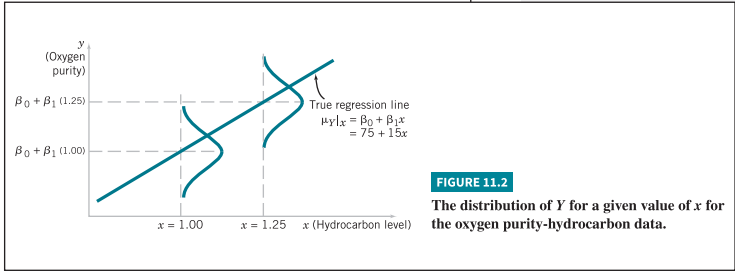


FIGURE 11.2 The distribution of Y for a given value of x for the oxygen purity-hydrocarbon data.

Computer Output

Example throughout the book, use computer output to illustrate the role of modern statistical software.

Stem-and-leaf of Strength

N = 80	Stem	Unit = 1.0
1	7	6
2	8	7
3	9	7
5	10	1 5
8	11	0 5 8
11	12	0 1 3
17	13	1 3 3 4 5 5
25	14	1 2 3 5 6 8 9 9
37	15	0 0 1 3 4 4 6 7 8 8 8 8
(10)	16	0 0 0 3 3 5 7 7 8 9
33	17	0 1 1 2 4 4 5 6 6 8
23	18	0 0 1 1 3 4 6
16	19	0 3 4 6 9 9
10	20	0 1 7 8
6	21	8
5	22	1 8 9
2	23	7
1	24	5

FIGURE 6.5 A typical computer-generated stem-and-leaf diagram.

Example Problems

A set of example problems provides the student with detailed solutions and comments for interesting, real-world situations. Brief practical interpretations have been added in this edition.

EXAMPLE 10.1 | Paint Drying Time

A product developer is interested in reducing the drying time of a primer paint. Two formulations of the paint are tested; formulation 1 is the standard chemistry, and formulation 2 has a new drying ingredient that should reduce the drying time. From experience, it is known that the standard deviation of drying time is 8 minutes, and this inherent variability should be unaffected by the addition of the new ingredient. Ten specimens are painted with formulation 1, and another 10 specimens are painted with formulation 2; the 20 specimens are painted in random order. The two sample average drying times are $\bar{x}_1 = 121$ minutes and $\bar{x}_2 = 112$ minutes, respectively. What conclusions can the product developer draw about the effectiveness of the new ingredient, using $\alpha = 0.05$?

We apply the seven-step procedure to this problem as follows:

- Parameter of interest:** The quantity of interest is the difference in mean drying times, $\mu_1 - \mu_2$, and $\Delta_0 = 0$.
- Null hypothesis:** $H_0: \mu_1 - \mu_2 = 0$, or $H_0: \mu_1 = \mu_2$.
- Alternative hypothesis:** $H_1: \mu_1 > \mu_2$. We want to reject H_0 if the new ingredient reduces mean drying time.

4. Test statistic: The test statistic is

$$z_0 = \frac{\bar{x}_1 - \bar{x}_2 - 0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \quad \text{where } \sigma_1^2 = \sigma_2^2 = (8)^2 = 64 \text{ and } n_1 = n_2 = 10.$$

5. Reject H_0 if: Reject $H_0: \mu_1 = \mu_2$ if the P -value is less than 0.05.

6. Computations: Because $\bar{x}_1 = 121$ minutes and $\bar{x}_2 = 112$ minutes, the test statistic is

$$z_0 = \frac{121 - 112}{\sqrt{\frac{(8)^2}{10} + \frac{(8)^2}{10}}} = 2.52$$

7. Conclusion: Because $z_0 = 2.52$, the P -value is $P = 1 - \Phi(2.52) = 0.0059$, so we reject H_0 at the $\alpha = 0.05$ level.

Practical Interpretation: We conclude that adding the new ingredient to the paint significantly reduces the drying time. This is a strong conclusion.

Exercises

Each chapter has an extensive collection of exercises, including **end-of-section exercises** that emphasize the material in that section, **supplemental exercises** at the end of the chapter that cover the scope of chapter topics and require the student to make a decision about the approach they will use to solve the problem, and **mind-expanding exercises** that often require the student to extend the text material somewhat or to apply it in a novel situation. Answers are provided to most odd-numbered exercises in Appendix C in the text, and the *WileyPLUS* online learning environment includes for students complete detailed solutions to selected exercises.

Important Terms and Concepts

At the end of each chapter is a list of important terms and concepts for an easy self-check and study tool.

Exercises for Section 2.2

WP

Problem available in WileyPLUS at instructor's discretion.

WP GO Tutorial

Tutoring problem available in WileyPLUS at instructor's discretion

SS

Student solution available in WileyPLUS.

VS

Student video solution available in WileyPLUS.

2.2.1 WP

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Supplemental Exercises for Chapter 2

WP

Problem available in WileyPLUS at instructor's discretion.

WP GO Tutorial

Tutoring problem available in WileyPLUS at instructor's discretion

SS

Student solution available in WileyPLUS.

VS

Student video solution available in WileyPLUS.

2.54 WP

Samples of laboratory glass are in small, light packaging or heavy, large packaging. Suppose that 2% and 1%, respectively, of the sample shipped in small and large packages, respectively, break during transit. If 60% of the samples are shipped in large packages and 40% are shipped in small packages, what proportion of samples break during shipment?

2.55 WP SS

Samples of a cast aluminum part are classified on the basis of surface finish (in microinches) and edge finish. The results of 100 parts are summarized as follows:
selected randomly, without replacement, from the lot of 40. Let A be the event that the first casting selected is from the local supplier, and let B denote the event that the second casting is selected from the local supplier. Determine:
a. $P(A)$ b. $P(B|A)$ c. $P(A \cap B)$ d. $P(A \cup B)$
Suppose that 3 castings are selected at random, without replacement, from the lot of 40. In addition to the definitions of events A and B , let C denote the event that the third casting selected is from the local supplier. Determine:
e. $P(A \cap B \cap C)$ f. $P(A \cap B \cap C')$
2.57 If A , B , and C are mutually exclusive events, is it possible for $P(A) = 0.3$, $P(B) = 0.4$, and $P(C) = 0.5$? Why or why not?
2.58 WP Incoming calls to a customer service center are classified as complaints (75% of calls) or requests for information (25% of calls). Of the complaints, 40% deal with computer equipment that does not respond and 57% deal with incomplete software installation; in the remaining 3% of complaints, the user has improperly followed the installation instructions. The requests for

2.2.9

In the laboratory analysis of samples from a chemical process, five samples from the process are analyzed daily. In addition, a control sample is analyzed twice each day to check the calibration of the laboratory instruments.
a. How many different sequences of process and control

Edge Finish

	Excellent	Good
Surface finish	80	2
Good	10	8

Let A denote the event that a sample has excellent surface finish, and let B denote the event that a sample has excellent edge finish. If a part is selected at random, determine the following probabilities:
a. $P(A)$ b. $P(B)$ c. $P(A')$
d. $P(A \cap B)$ e. $P(A \cup B)$ f. $P(A' \cup B)$
2.56 WP A lot contains 15 castings from a local supplier and 25 castings from a supplier in the next state. Two castings are Use a tree diagram to determine the following:
a. $P(B)$ b. $P(A|B)$ c. $P(A|B')$
d. $P(A \cap B)$ e. $P(A \cap B')$ f. $P(A)$
g. What is the probability that the useful life of a laser exceeds five years?
h. What is the probability that a laser that failed before five years came from a product used for backup?
2.511 WP SS A congested computer network has a 0.002 probability of losing a data packet, and packet losses are independent events. A lost packet must be resent.
a. What is the probability that an e-mail message with 100 packets will need to be resent?
b. What is the probability that an e-mail message with 3 packets will need exactly 1 to be resent?
c. If 10 e-mail messages are sent, each with 100 packets, what is the probability that at least 1 message will need some packets to be resent?
2.512 WP GO Tutorial An electronic storage device uses an error recovery procedure that requires an immediate satisfactory read-back of any written data. If the read-back is not successful after eliminated tion of the 8. Assume ity that an eptable for

Meticulous Ensuring of Accuracy

There is always room for improvement in this area for richly mathematical texts. We appreciate the input of sixth edition adopters and made achieving the greatest possible accuracy of every element of the seventh edition a very high-priority objective. We have been rigorous in reviewing the retained and new content in the text and reviewing and testing the gradable exercises in the WileyPLUS course. We thank the colleagues whom Wiley engaged to execute these activities: (1) the entire text, including all exposition, examples, exercises, and answers have been read and checked for accuracy, including calculations; (2) answers and solutions to all exercises have been checked via independent solution of problems; (3) all questions and exercises in the WileyPLUS course have been checked with a focus on verifying the answers, correcting any conflicting input instructions, and improving the precision of the numerical answers.

Increased Support for Student Engagement

The rapidly increasing use of approaches like flipped classrooms and online course delivery has motivated us to enhance the features for student self-study, especially those that can better engage

students. The following new and expanded interactive features are available in both the WileyPLUS course and the interactive e-text:

- *Video-based examples and problem walk-throughs.* There are now over 100 video solutions and about 40 example videos.
- *Check Your Understanding* exercises are placed at the end of each appropriate section in the WileyPLUS e-text. These simple practice exercises allow students to test whether they comprehend the material they have just read. They can also be used, if the instructor prefers, to assess student readiness to apply concepts and techniques. The program contains over 600 Check Your Understanding questions.

Greater Selection of Assignable Exercises

The seventh edition contains over 1800 exercises, an increase of about 25%. Over 1100 of these exercises are available to the students and appear in the instructor desk copy, e-text version of the text, and about 750 of these exercises have been reserved for instructors in WileyPlus. The e-text and WileyPLUS versions have also added about 300 new exercises to the Check Your Understanding program that consists of about 600 questions. These questions are integrated into the appropriate point in the text exposition to provide students with the ability to test their comprehension of the material just presented before proceeding. In the WileyPLUS course the Check Your Understanding exercises are assignable, enabling instructors to assess student readiness.

Instructor-Only and Algorithmic Exercises for More Exercise Security

The wide and inexpensive availability of solutions to all published textbook problems has added a huge challenge for instructors looking to assess learning. Wiley is embarking on a comprehensive strategy to provide original and reconfigured problems that can decrease the proliferation of problem solutions on the Web. As a first step in this direction, the instructor evaluation/desk copy of this text includes about 750 problems that are *reserved for the instructor only—they do not appear in the student version*. These problems are provided in the instructor desk copy for easy reference. All of these reserved questions appear as assignable problems in WileyPLUS. About 40% of these problems are entirely brand new. The other 60% of these problems have been revised from earlier editions and are now only available as assignable problems. The WileyPLUS course also features *algorithmic homework problems*, machine-gradable problems in which each student has unique values for key variables.

Exercises Collected in the Back of the Book in Instructor Desk Copy

Many instructors have requested that we curtail the practice of frequent revisions of the text. At the same time, the demand for fresh and new problems among instructors is quite high. The exercises in the seventh edition are collected in the back of the instructor desk copy as well as the student e-text and WileyPLUS e-text. We have done this in effort to be able to generate entirely new problem collections that can be adopted without altering the contents of the book. We expect that revisions going forward will be confined largely to expansion of learning aids in the digital forms of the text and updates of the problems and to elongate the life of the print text.

STUDENT RESOURCES

- **Data Sets** Data sets for all examples and exercises in the text. Visit the student section of the book Web site at www.wiley.com/college/montgomery to access these materials.

INSTRUCTOR RESOURCES

The following resources are available only to instructors who adopt the text:

- **Solutions Manual** All solutions to the exercises in the text.
- **Data Sets** Data sets for all examples and exercises in the text.
- **Image Gallery of Text Figures**
- **Section on Logistic Regression**

These instructor-only resources are password-protected. Visit the instructor section of the book Web site at www.wiley.com/college/montgomery to register for a password to access these materials.

COMPUTER SOFTWARE

We have used several different packages, including Excel, to demonstrate computer usage. Minitab can be used for most exercises. A student version of Minitab is available as an option to purchase in a set with this text. Student versions of software often do not have all the functionality that full versions do. Consequently, student versions may not support all the concepts presented in this text. If you would like to adopt for your course the set of this text with the student version of Minitab, please contact your local Wiley representative at www.wiley.com/college/rep.

Alternatively, students may find information about how to purchase the professional version of the software for academic use at www.minitab.com.

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COURSE SYLLABUS SUGGESTIONS

This is a very flexible textbook because instructors' ideas about what should be in a first course on statistics for engineers vary widely, as do the abilities of different groups of students. Therefore, we hesitate to give too much advice, but will explain how we use the book.

We believe that a first course in statistics for engineers should be primarily an applied statistics course, not a probability course. In our one-semester course we cover all of Chapter 1 (in one or two lectures); overview the material on probability, putting most of the emphasis on the normal distribution (six to eight lectures); discuss most of Chapters 6 through 10 on confidence intervals and tests (twelve to fourteen lectures); introduce regression models in Chapter 11 (four lectures); give an introduction to the design of experiments from Chapters 13 and 14 (six lectures); and present the basic concepts of statistical process control, including the Shewhart control chart from Chapter 15 (four lectures). This leaves about three to four periods for exams and review. Let us emphasize that the purpose of this course is to introduce engineers to how statistics can be used to solve real-world engineering problems, not to weed out the less mathematically gifted students. This course is not the “baby math-stat” course that is all too often given to engineers.

If a second semester is available, it is possible to cover the entire book, including much of the supplemental material, if appropriate for the audience. It would also be possible to assign and work many of the homework problems in class to reinforce the understanding of the concepts. Obviously, multiple regression and more design of experiments would be major topics in a second course.

USING THE COMPUTER

In practice, engineers use computers to apply statistical methods to solve problems. Therefore, we strongly recommend that the computer be integrated into the class. Throughout the book we have presented typical example of the output that can be obtained with modern statistical software. In teaching, we have used a variety of software packages, including Minitab, Statgraphics, JMP, and Statistica. We did not clutter up the book with operational details of these different packages because how the instructor integrates the software into the class is ultimately more important than which package is used. All text data are available in electronic form on the textbook Web site. We have marked these problems with a special icon in the margin.

In our own classrooms, we use the computer in almost every lecture and demonstrate how the technique is implemented in software as soon as it is discussed in the lecture. Many educational institutions have file licenses for statistics software and students can access it for class use. Student versions of many statistical software packages are available at low cost, and students can either purchase their own copy or use the products available through the institution. We have found that greatly improves the pace of the course and student understanding of the material.

Users should be aware that final answers may differ slightly due to different numerical precision and rounding protocols among softwares.

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DOUGLAS C. MONTGOMERY

GEORGE C. RUNGER

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Continuous Random Variables and Probability Distributions

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LEARNING OBJECTIVES

After careful study of this chapter, you should be able to do the following:

1. Determine probabilities from probability density functions
2. Determine probabilities from cumulative distribution functions and cumulative distribution functions from probability density functions, and the reverse
3. Calculate means and variances for continuous random variables
4. Understand the assumptions for some common continuous probability distributions
5. Calculate probabilities, determine means and variances for some common continuous probability distributions
6. Approximate probabilities for binomial and Poisson distributions

CHAPTER OUTLINE

- 4.1 Probability Distributions and Probability Density Functions**
- 4.2 Cumulative Distribution Functions**
- 4.3 Mean and Variance of a Continuous Random Variable**
- 4.4 Continuous Uniform Distribution**
- 4.5 Normal Distribution**
- 4.6 Normal Approximation to the Binomial and Poisson Distributions**
- 4.7 Exponential Distribution**
- 4.8 Erlang and Gamma Distributions**
- 4.9 Weibull Distribution**
- 4.10 Lognormal Distribution**
- 4.11 Beta Distribution**

The kinetic theory of gases provides a link between statistics and physical phenomena. Physicist James Maxwell used some basic assumptions to determine the distribution of molecular velocity in a gas at equilibrium. As a result of molecular collisions, all directions of rebound are equally likely. From this concept, he assumed equal probabilities for velocities in all the x , y , and z directions and independence of these components of velocity. This alone is sufficient to show that the probability distribution of the velocity in a particular direction x is the continuous probability distribution known as the normal distribution. This fundamental probability distribution can be derived from other approaches (such as the central limit theorem, to be discussed in a later chapter), but the kinetic theory may be the most parsimonious. This role for the normal distribution illustrates one example of the importance of continuous probability distributions within science and engineering.

4.1 Probability Distributions and Probability Density Functions

Suppose that a dimensional length is measured on a manufactured part selected from a day's production. In practice, there can be small variations in the measurements due to many causes, such as vibrations, temperature fluctuations, operator differences, calibrations, cutting tool wear, bearing wear, and raw material changes. In an experiment such as this, the measurement is naturally represented as a random variable X , and it is reasonable to model the range of possible values of X with an interval of real numbers. Recall from Chapter 2 that a **continuous random variable** is a random variable with an interval (either finite or infinite) of real numbers for its range. The model provides for any precision in length measurements.

Because the number of possible values of X is uncountably infinite, X has a distinctly different distribution from the discrete random variables studied previously. But as in the discrete case, many physical systems can be modeled by the same or similar continuous random variables. These random variables are described, and example computations of probabilities, means, and variances are provided in the sections of this chapter.

Density functions are commonly used in engineering to describe physical systems. For example, consider the density of a loading on a long, thin beam as shown in Figure 4.1. For any point x along the beam, the density can be described by a function (in grams/cm). Intervals with large loadings correspond to large values for the function. The total loading between points a and b is determined as the integral of the density function from a to b . This integral is the area under the density function over this interval, and it can be loosely interpreted as the sum of all the loadings over this interval.

Similarly, a **probability density function** $f(x)$ can be used to describe the probability distribution of a continuous random variable X . If an interval is likely to contain a value for X , its probability is large and it corresponds to large values for $f(x)$. The probability that X is between a and b is determined as the integral of $f(x)$ from a to b . See Figure 4.2.

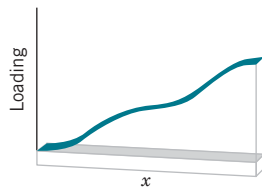


FIGURE 4.1

Density function of a loading on a long, thin beam.

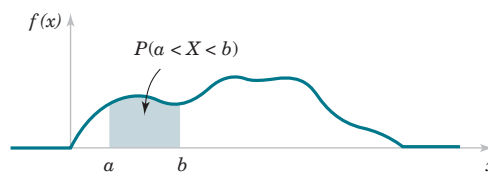


FIGURE 4.2

Probability determined from the area under $f(x)$.

Probability Density Function

For a continuous random variable X , a **probability density function** is a function such that

- (1) $f(x) \geq 0$
- (2) $\int_{-\infty}^{\infty} f(x) dx = 1$
- (3) $P(a \leq X \leq b) = \int_a^b f(x) dx = \text{area under } f(x) \text{ from } a \text{ to } b \text{ for any } a \text{ and } b$ (4.1)

A probability density function provides a simple description of the probabilities associated with a random variable. As long as $f(x)$ is nonnegative and $\int_{-\infty}^{\infty} f(x) dx = 1$, $0 \leq P(a < X < b) \leq 1$ so that the probabilities are properly restricted. A probability density function is zero for x values that cannot occur, and it is assumed to be zero wherever it is not specifically defined.

A **histogram** is an approximation to a probability density function. See Figure 4.3. For each interval of the histogram, the area of the bar equals the relative frequency (proportion) of the measurements in the interval. The relative frequency is an estimate of the probability that a measurement falls in the interval. Similarly, the area under $f(x)$ over any interval equals the true probability that a measurement falls in the interval.

The important point is that $f(x)$ is used to calculate an area that represents the probability that X assumes a value in $[a, b]$. For the current measurement example, the probability that X results in $[14 \text{ mA}, 15 \text{ mA}]$ is the integral of the probability density function of X over this interval. The probability that X results in $[14.5 \text{ mA}, 14.6 \text{ mA}]$ is the integral of the same function, $f(x)$, over the smaller interval. By appropriate choice of the shape of $f(x)$, we can represent the probabilities associated with any continuous random variable X . The shape of $f(x)$ determines how the probability that X assumes a value in $[14.5 \text{ mA}, 14.6 \text{ mA}]$ compares to the probability of any other interval of equal or different length.

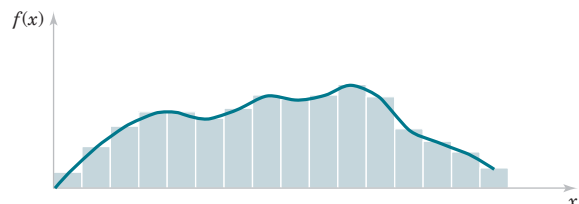
For the density function of a loading on a long, thin beam, because every point has zero width, the loading at any point is zero. Similarly, for a continuous random variable X and any value x ,

$$P(X = x) = 0$$

Based on this result, it might appear that our model of a continuous random variable is useless. However, in practice, when a particular current measurement, such as 14.47 milliamperes, is observed, this result can be interpreted as the rounded value of a current measurement that is actually in a range such as $14.465 \leq x \leq 14.475$. Therefore, the probability that the rounded value 14.47 is observed as the value for X is the probability that X assumes a value in the interval $[14.465, 14.475]$, which is not zero. Similarly, because each point has zero probability, one need not distinguish between inequalities such as $<$ or $\leq$ for continuous random variables.

FIGURE 4.3

Histogram approximates a probability density function.



If X is a *continuous random variable*, for any x_1 and x_2 ,

$$P(x_1 \leq X \leq x_2) = P(x_1 < X \leq x_2) = P(x_1 \leq X < x_2) = P(x_1 < X < x_2) \quad (4.2)$$

EXAMPLE 4.1 | Electric Current

Let the continuous random variable X denote the current measured in a thin copper wire in milliamperes. Assume that the range of X is $[4.9, 5.1]$ mA, and assume that the probability density function of X is $f(x) = 5$ for $4.9 \leq x \leq 5.1$. What is the probability that a current measurement is less than 5 milliamperes?

The probability density function is shown in Figure 4.4. It is assumed that $f(x) = 0$ wherever it is not specifically

defined. The shaded area in Figure 4.4 indicates the probability.

$$P(X < 5) = \int_{4.9}^5 f(x) dx = \int_{4.9}^5 5 dx = 0.5$$

As another example,

$$P(4.95 < X < 5.1) = \int_{4.95}^{5.1} f(x) dx = 0.75$$

EXAMPLE 4.2 | Hole Diameter

Let the continuous random variable X denote the diameter of a hole drilled in a sheet metal component. The target diameter is 12.5 millimeters. Most random disturbances to the process result in larger diameters. Historical data show that the distribution of X can be modeled by a probability density function $f(x) = 20e^{-20(x-12.5)}$, for $x \geq 12.5$.

If a part with a diameter greater than 12.60 mm is scrapped, what proportion of parts is scrapped? The density function and the requested probability are shown in Figure 4.5. A part is scrapped if $X > 12.60$. Now,

$$\begin{aligned} P(X > 12.60) &= \int_{12.6}^{\infty} f(x) dx = \int_{12.6}^{\infty} 20e^{-20(x-12.5)} dx \\ &= -e^{-20(x-12.5)} \Big|_{12.6}^{\infty} = 0.135 \end{aligned}$$

What proportion of parts is between 12.5 and 12.6 millimeters? Now,

$$\begin{aligned} P(12.5 < X < 12.6) &= \int_{12.5}^{12.6} f(x) dx \\ &= -e^{-20(x-12.5)} \Big|_{12.5}^{12.6} = 0.865 \end{aligned}$$

Because the total area under $f(x)$ equals 1, we can also calculate $P(12.5 < X < 12.6) = 1 - P(X > 12.6) = 1 - 0.135 = 0.865$.

Practical Interpretation: Because 0.135 is the proportion of parts with diameters greater than 12.60 mm, a large proportion of parts is scrapped. Process improvements are needed to increase the proportion of parts with dimensions near 12.50 mm.

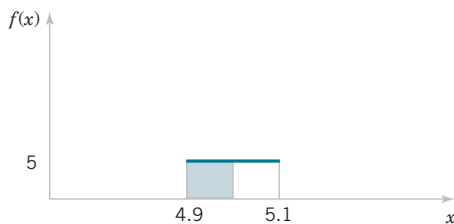


FIGURE 4.4

Probability density function for Example 4.1.

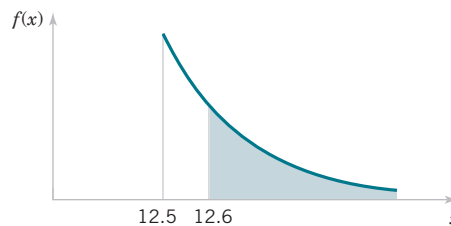


FIGURE 4.5

Probability density function for Example 4.2.

4.2

Cumulative Distribution Functions

An alternative method to describe the distribution of a discrete random variable can also be used for continuous random variables.

Cumulative Distribution Function

The **cumulative distribution function** of a continuous random variable X is

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(u) du \quad (4.3)$$

for $-\infty < x < \infty$.

The cumulative distribution function is defined for all real numbers.

EXAMPLE 4.3 | Electric Current

For the copper current measurement in Example 4.1, the cumulative distribution function of the random variable X consists of three expressions. If $x < 4.9$, $f(x) = 0$. Therefore,

$$F(x) = 0, \quad \text{for } x < 4.9$$

and

$$F(x) = \int_{4.9}^x f(u) du = 5x - 24.5, \quad \text{for } 4.9 \leq x < 5.1$$

Finally,

$$F(x) = \int_{4.9}^x f(u) du = 1, \quad \text{for } 5.1 \leq x$$

Therefore,

$$F(x) = \begin{cases} 0 & x < 4.9 \\ 5x - 24.5 & 4.9 \leq x < 5.1 \\ 1 & 5.1 \leq x \end{cases}$$

The plot of $F(x)$ is shown in Figure 4.6.

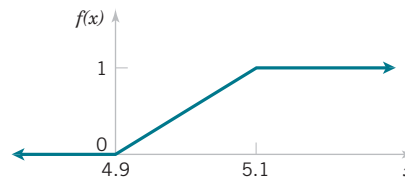


FIGURE 4.6

Cumulative distribution function for Example 4.3.

Notice that in the definition of $F(x)$, any $<$ can be changed to $\leq$ and vice versa. That is, in Example 4.3 $F(x)$ can be defined as either $5x - 24.5$ or 0 at the end-point $x = 4.9$, and $F(x)$ can be defined as either $5x - 24.5$ or 1 at the end-point $x = 5.1$. In other words, $F(x)$ is a continuous function. For a discrete random variable, $F(x)$ is not a continuous function. Sometimes a continuous random variable is defined as one that has a continuous cumulative distribution function.

The probability density function of a continuous random variable can be determined from the cumulative distribution function by differentiating. The fundamental theorem of calculus states that

$$\frac{d}{dx} \int_{-\infty}^x f(u) du = f(x)$$

Probability Density Function from the Cumulative Distribution Function

Given $F(x)$,

$$f(x) = \frac{dF(x)}{dx}$$

as long as the derivative exists.

EXAMPLE 4.4 | Reaction Time

The time until a chemical reaction is complete (in milliseconds) is approximated by the cumulative distribution function

$$F(x) = \begin{cases} 0 & x < 0 \\ 1 - e^{-0.01x} & 0 \leq x \end{cases}$$

Determine the probability density function of X . What proportion of reactions is complete within 200 milliseconds?

Using the result that the probability density function is the derivative of $F(x)$, we obtain

$$f(x) = \begin{cases} 0 & x < 0 \\ 0.01e^{-0.01x} & 0 \leq x \end{cases}$$

The probability that a reaction completes within 200 milliseconds is

$$P(X < 200) = F(200) = 1 - e^{-2} = 0.8647$$

4.3 Mean and Variance of a Continuous Random Variable

The mean and variance can also be defined for a continuous random variable. Integration replaces summation in the discrete definitions. If a probability density function is viewed as a loading on a beam as in Figure 4.1, the mean is the balance point.

Mean and Variance

Suppose that X is a continuous random variable with probability density function $f(x)$. The **mean** or **expected value** of X , denoted as μ or $E(X)$, is

$$\mu = E(X) = \int_{-\infty}^{\infty} xf(x)dx \quad (4.4)$$

The **variance** of X , denoted as $V(X)$ or σ^2 , is

$$\sigma^2 = V(X) = \int_{-\infty}^{\infty} (x - \mu)^2 f(x)dx = \int_{-\infty}^{\infty} x^2 f(x)dx - \mu^2$$

The **standard deviation** of X is $\sigma = \sqrt{\sigma^2}$.

The equivalence of the two formulas for variance can be derived from the same approach used for discrete random variables.

EXAMPLE 4.5 | Electric Current

For the copper current measurement in Example 4.1, the mean of X is

$$E(X) = \int_{4.9}^{5.1} xf(x)dx = \frac{5x^2}{2} \Big|_{4.9}^{5.1} = 5$$

The variance of X is

$$V(X) = \int_{4.9}^{5.1} (x - 5)^2 f(x)dx = \frac{5(x - 5)^3}{3} \Big|_{4.9}^{5.1} = 0.0033$$

The expected value of a function $h(X)$ of a continuous random variable is also defined in a straightforward manner.

Expected Value of a Function of a Continuous Random Variable

If X is a continuous random variable with probability density function $f(x)$,

$$E[h(X)] = \int_{-\infty}^{\infty} h(x)f(x)dx \quad (4.5)$$

In the special case that $h(X) = aX + b$ for any constants a and b , $E[h(X)] = aE(X) + b$. This can be shown from the properties of integrals.

EXAMPLE 4.6

In Example 4.1, X is the current measured in milliamperes. What is the expected value of power when the resistance is 100 ohms? Use the result that power in watts $P = 10^{-6}RI^2$, where I is the current in milliamperes and

R is the resistance in ohms. Now, $h(X) = (10^{-6})100X^2$. Therefore,

$$E[h(X)] = 10^{-4} \int_{4.9}^{5.1} 5x^2 dx = 0.0001 \frac{x^3}{3} \Big|_{4.9}^{5.1} = 0.0025 \text{ watts}$$

4.4**Continuous Uniform Distribution**

The simplest continuous distribution is analogous to its discrete counterpart.

Continuous Uniform Distribution

A continuous random variable X with probability density function

$$f(x) = 1/(b-a), \quad a \leq x \leq b \quad (4.6)$$

is a **continuous uniform random variable**.

The probability density function of a continuous uniform random variable is shown in Figure 4.7. The mean of the continuous uniform random variable X is

$$E(X) = \int_a^b \frac{x}{b-a} dx = \frac{0.5x^2}{b-a} \Big|_a^b = \frac{(a+b)}{2}$$

The variance of X is

$$V(X) = \int_a^b \frac{\left(x - \frac{(a+b)}{2}\right)^2}{b-a} dx = \frac{\left(x - \frac{(a+b)}{2}\right)^3}{3(b-a)} \Big|_a^b = \frac{(b-a)^2}{12}$$

These results are summarized as follows.

**FIGURE 4.7**

Continuous uniform probability density function.

Mean and Variance

If X is a continuous uniform random variable over $a \leq x \leq b$,

$$\mu = E(X) = \frac{a+b}{2} \quad \text{and} \quad \sigma^2 = V(X) = \frac{(b-a)^2}{12} \quad (4.7)$$

EXAMPLE 4.7 | Uniform Current

In Example 4.1, the random variable X has a continuous uniform distribution on $[4.9, 5.1]$. The probability density function of X is $f(x) = 5$, $4.9 \leq x \leq 5.1$.

What is the probability that a measurement of current is between 4.95 and 5.0 milliamperes? The requested probability is shown as the shaded area in Figure 4.8.

$$P(4.95 < x < 5.0) = \int_{4.95}^{5.0} f(x)dx = 5(0.05) = 0.25$$

The mean and variance formulas can be applied with $a = 4.9$ and $b = 5.1$. Therefore,

$$E(X) = 5 \text{ mA} \quad \text{and} \quad V(X) = 0.2^2/12 = 0.0033 \text{ mA}^2$$

Consequently, the standard deviation of X is 0.0577 mA.

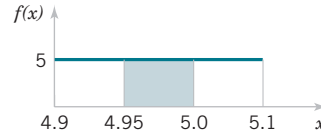


FIGURE 4.8

Probability for Example 4.7.

The cumulative distribution function of a continuous uniform random variable is obtained by integration. If $a < x < b$,

$$F(x) = \int_a^x \frac{1}{b-a} du = \frac{x-a}{b-a}$$

Therefore, the complete description of the cumulative distribution function of a continuous uniform random variable is

$$F(x) = \begin{cases} 0 & x < a \\ \frac{x-a}{b-a} & a \leq x < b \\ 1 & b \leq x \end{cases}$$

An example of $F(x)$ for a continuous uniform random variable is shown in Figure 4.6.

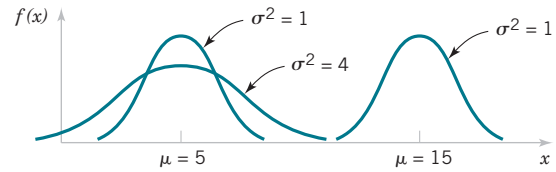
4.5 Normal Distribution

Undoubtedly, the most widely used model for a continuous measurement is a **normal random variable**. Whenever a random experiment is replicated, the random variable that equals the average (or total) result over the replicates tends to have a normal distribution as the number of replicates becomes large. De Moivre presented this fundamental result, known as the **central limit theorem**, in 1733. Unfortunately, his work was lost for some time, and Gauss independently developed a normal distribution nearly 100 years later. Although De Moivre was later credited with the derivation, a normal distribution is also referred to as a **Gaussian distribution**.

When do we average (or total) results? Almost always. For example, an automotive engineer may plan a study to average pull-off force measurements from several connectors. If we assume that each measurement results from a replicate of a random experiment, the normal distribution can be used to make approximate conclusions about this average. These conclusions are the primary topics in the subsequent chapters of this book.

FIGURE 4.9

Normal probability density functions for selected values of the parameters μ and σ^2 .



Furthermore, sometimes the central limit theorem is less obvious. For example, assume that the deviation (or error) in the length of a machined part is the sum of a large number of infinitesimal effects, such as temperature and humidity drifts, vibrations, cutting angle variations, cutting tool wear, bearing wear, rotational speed variations, mounting and fixture variations, variations in numerous raw material characteristics, and variations in levels of contamination. If the component errors are independent and equally likely to be positive or negative, the total error can be shown to have an approximate normal distribution. Furthermore, the normal distribution arises in the study of numerous basic physical phenomena. For example, physicist James Maxwell developed a normal distribution from simple assumptions regarding the velocities of molecules. Our objective now is to calculate probabilities for a normal random variable. The central limit theorem is stated more carefully in Chapter 5.

Figure 4.9 illustrates several normal probability density functions with selected values of the mean μ and the variance σ^2 . Each has the characteristic symmetric bell-shaped curve, but the centers and dispersions differ. The following definition provides the formula for normal probability density functions.

Normal Distribution

A random variable X with probability density function

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad -\infty < x < \infty \quad (4.8)$$

is a **normal random variable** with parameters μ where $-\infty < \mu < \infty$ and $\sigma > 0$. Also,

$$E(X) = \mu \quad \text{and} \quad V(X) = \sigma^2 \quad (4.9)$$

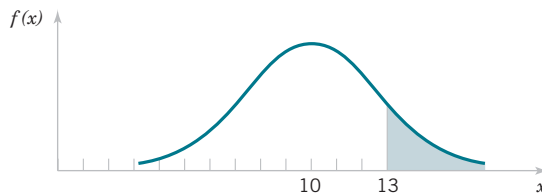
and the notation $N(\mu, \sigma^2)$ is used to denote the distribution.

The mean and variance of X are shown to equal μ and σ^2 , respectively, in an online exercise in Chapter 5.

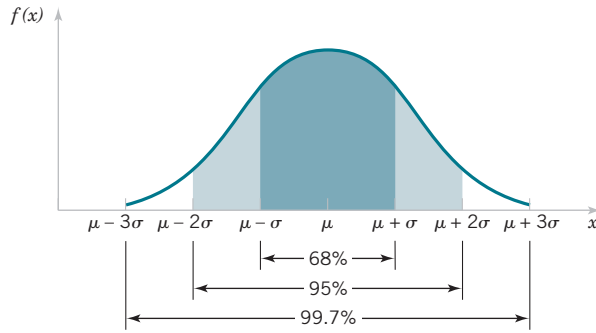
EXAMPLE 4.8

Assume that the current measurements in a strip of wire follow a normal distribution with a mean of 10 milliamperes and a variance of 4 (milliamperes)². What is the probability that a measurement exceeds 13 milliamperes?

Let X denote the current in milliamperes. The requested probability can be represented as $P(X > 13)$. This probability is shown as the shaded area under the normal probability density function in Figure 4.10. Unfortunately, there is no closed-form expression for the integral of a normal probability density function, and probabilities based on the normal distribution are typically found numerically or from a table (that we introduce soon).

**FIGURE 4.10**

Probability that $X > 13$ for a normal random variable with $\mu = 10$ and $\sigma^2 = 4$.

**FIGURE 4.11**

Probabilities associated with a normal distribution.

The following equations and Figure 4.11 summarize some useful results concerning a normal distribution. For any normal random variable,

$$P(\mu - \sigma < X < \mu + \sigma) = 0.6827$$

$$P(\mu - 2\sigma < X < \mu + 2\sigma) = 0.9545$$

$$P(\mu - 3\sigma < X < \mu + 3\sigma) = 0.9973$$

Also, from the symmetry of $P(X < \mu) = P(X > \mu) = 0.5$. Because $f(x)$ is positive for all x , this model assigns some probability to each interval of the real line. However, the probability density function decreases as x moves farther from μ . Consequently, the probability that a measurement falls far from μ is small, and at some distance from μ , the probability of an interval can be approximated as zero.

The area under a normal probability density function beyond 3σ from the mean is quite small. This fact is convenient for quick, rough sketches of a normal probability density function. The sketches help us determine probabilities. Because more than 0.9973 of the probability of a normal distribution is within the interval $(\mu - 3\sigma, \mu + 3\sigma)$, 6σ is often referred to as the **width** of a normal distribution. Advanced integration methods can be used to show that the area under the normal probability density function from $-\infty < x < \infty$ is 1.

Standard Normal Random Variable

A normal random variable with

$$\mu = 0 \quad \text{and} \quad \sigma^2 = 1$$

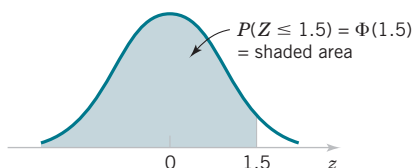
is called a **standard normal random variable** and is denoted as Z . The cumulative distribution function of a standard normal random variable is denoted as

$$\Phi(z) = P(Z \leq z)$$

Appendix Table III provides cumulative probabilities for a standard normal random variable. Cumulative distribution functions for normal random variables are also widely available in computer packages. They can be used in the same manner as Appendix Table III to obtain probabilities for these random variables. The use of Table III is illustrated by the following example.

EXAMPLE 4.9 | Standard Normal Distribution

Assume that Z is a standard normal random variable. Appendix Table III provides probabilities of the form $\Phi(z) = P(Z \leq z)$. The use of Table III to find $P(Z \leq 1.5)$ is illustrated in Figure 4.12. Read down the z column to the row that equals 1.5. The probability is read from the adjacent column, labeled 0.00, to be 0.93319.



z	0.00	0.01	0.02	0.03
0	0.50000	0.50399	0.50800	0.51197
$\vdots$		$\vdots$		
1.5	0.93319	0.93448	0.93574	0.93699

FIGURE 4.12

Standard normal probability density function.

Probabilities that are not of the form $P(Z \leq z)$ are found by using the basic rules of probability and the symmetry of the normal distribution along with Appendix Table III. The following examples illustrate the method.

EXAMPLE 4.10 | Normal Distribution Calculations

The following calculations are shown pictorially in Figure 4.13. In practice, a probability is often rounded to one or two significant digits.

1. $P(Z > 1.26) = 1 - P(Z \leq 1.26) = 1 - 0.89616 = 0.10384$.
2. $P(Z < -0.86) = 0.19490$.
3. $P(Z > -1.37) = P(Z < 1.37) = 0.91465$.
4. $P(-1.25 < Z < 0.37)$. This probability can be found from the difference of two areas, $P(Z < 0.37) - P(Z < -1.25)$. Now,

$$P(Z < 0.37) = 0.64431$$

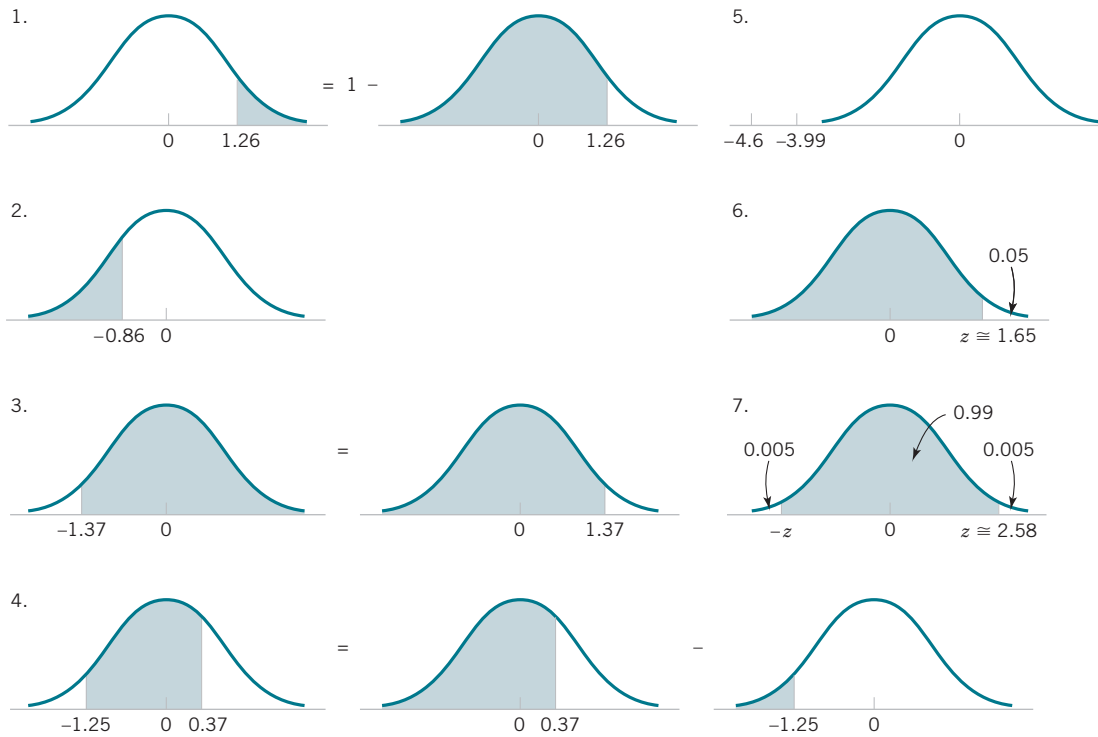
and

$$P(Z < -1.25) = 0.10565$$

Therefore,

$$\begin{aligned} P(-1.25 < Z < 0.37) &= 0.64431 - 0.10565 \\ &= 0.53866 \end{aligned}$$

5. $P(Z \leq -4.6)$ cannot be found exactly from Appendix Table III. However, the last entry in the table can be used to find that $P(Z \leq -3.99) = 0.00003$. Because $P(Z \leq -4.6) < P(Z \leq -3.99)$, $P(Z \leq -4.6)$ is nearly zero.
6. Find the value z such that $P(Z > z) = 0.05$. This probability expression can be written as $P(Z \leq z) = 0.95$. Now Table III is used in reverse. We search through the probabilities to find the value that corresponds to 0.95. The solution is illustrated in Figure 4.13. We do not find 0.95 exactly; the nearest value is 0.95053, corresponding to $z = 1.65$.
7. Find the value of z such that $P(-z < Z < z) = 0.99$. Because of the symmetry of the normal distribution, if the area of the shaded region in Figure 4.13(7) is to equal 0.99, the area in each tail of the distribution must equal 0.005. Therefore, the value for z corresponds to a probability of 0.995 in Table III. The nearest probability in Table III is 0.99506 when $z = 2.58$.

**FIGURE 4.13**

Graphical displays for standard normal distributions.

The cases in Example 4.10 show how to calculate probabilities for standard normal random variables. To use the same approach for an arbitrary normal random variable would require the availability of a separate table for every possible pair of values for μ and σ . Fortunately, all normal probability distributions are related algebraically, and Appendix Table III can be used to find the probabilities associated with an arbitrary normal random variable by first using a simple transformation.

Standardizing a Normal Random Variable

If X is a normal random variable with $E(X) = \mu$ and $V(X) = \sigma^2$, the random variable

$$Z = \frac{X - \mu}{\sigma} \quad (4.10)$$

is a normal random variable with $E(Z) = 0$ and $V(Z) = 1$. That is, Z is a standard normal random variable.

Creating a new random variable by this transformation is referred to as **standardizing**. The random variable Z represents the distance of X from its mean in terms of standard

deviations. It is the key step to calculating a probability for an arbitrary normal random variable.

EXAMPLE 4.11 | Normally Distributed Current

Suppose that the current measurements in a strip of wire are assumed to follow a normal distribution with a mean of 10 milliamperes and a variance of 4 (milliamperes)². What is the probability that a measurement exceeds 13 milliamperes?

Let X denote the current in milliamperes. The requested probability can be represented as $P(X > 13)$. Let $Z = (X - 10)/2$. The relationship between the several values of X and the transformed values of Z are shown in Figure 4.14. We note that $X > 13$ corresponds to $Z > 1.5$. Therefore, from Appendix Table III,

$$\begin{aligned} P(X > 13) &= P(Z > 1.5) = 1 - P(Z \leq 1.5) \\ &= 1 - 0.93319 = 0.06681 \end{aligned}$$

Rather than using Figure 4.14, the probability can be found from the inequality $X > 13$. That is,

$$\begin{aligned} P(X > 13) &= P\left(\frac{X - 10}{2} > \frac{13 - 10}{2}\right) \\ &= P(Z > 1.5) = 0.06681 \end{aligned}$$

Practical Interpretation: Probabilities for any normal random variable can be computed with a simple transform to a standard normal random variable.

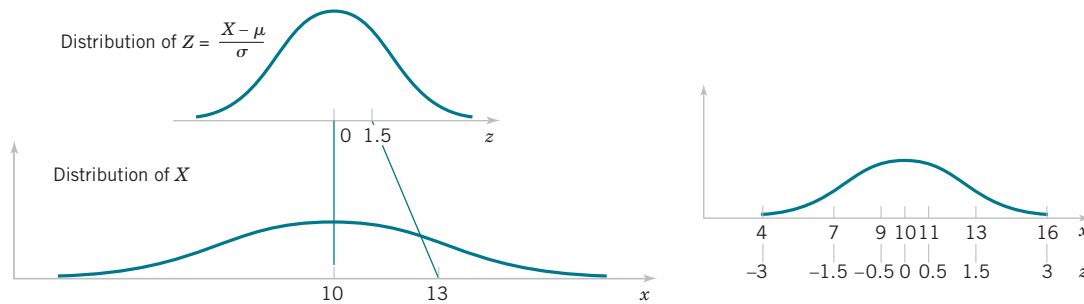


FIGURE 4.14

Standardizing a normal random variable.

In Example 4.11, the value 13 is transformed to 1.5 by standardizing, and 1.5 is often referred to as the **z-value** associated with a probability. The following summarizes the calculation of probabilities derived from normal random variables.

Standardizing to Calculate a Probability

Suppose that X is a normal random variable with mean μ and variance σ^2 . Then,

$$P(X \leq x) = P\left(\frac{X - \mu}{\sigma} \leq \frac{x - \mu}{\sigma}\right) = P(Z \leq z) \quad (4.11)$$

where Z is a **standard normal random variable**, and $z = \frac{(x - \mu)}{\sigma}$ is the **z-value** obtained by **standardizing** X . The probability is obtained by using Appendix Table III with $z = (x - \mu)/\sigma$.

EXAMPLE 4.12 | Normally Distributed Current

Continuing Example 4.11, what is the probability that a current measurement is between 9 and 11 milliamperes? From Figure 4.14, or by proceeding algebraically, we have

$$\begin{aligned} P(9 < X < 11) &= P\left(\frac{9-10}{2} < \frac{X-10}{2} < \frac{11-10}{2}\right) \\ &= P(-0.5 < Z < 0.5) \\ &= P(Z < 0.5) - P(Z < -0.5) \\ &= 0.69146 - 0.30854 = 0.38292 \end{aligned}$$

Determine the value for which the probability that a current measurement is less than this value is 0.98. The requested value is shown graphically in Figure 4.15. We need the value of x such that $P(X < x) = 0.98$. By standardizing, this probability expression can be written as

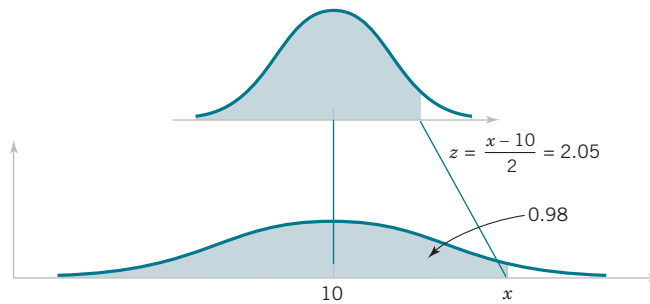
$$\begin{aligned} P(X < x) &= P\left(\frac{X-10}{2} < \frac{x-10}{2}\right) \\ &= P\left(Z < \frac{x-10}{2}\right) = 0.98 \end{aligned}$$

Appendix Table III is used to find the z -value such that $P(Z < z) = 0.98$. The nearest probability from Table III results in

$$P(Z < 2.06) = 0.980301$$

Therefore, $(x - 10)/2 = 2.06$, and the standardizing transformation is used in reverse to solve for x . The result is

$$x = 2(2.06) + 10 = 14.1 \text{ mA}$$

**FIGURE 4.15**

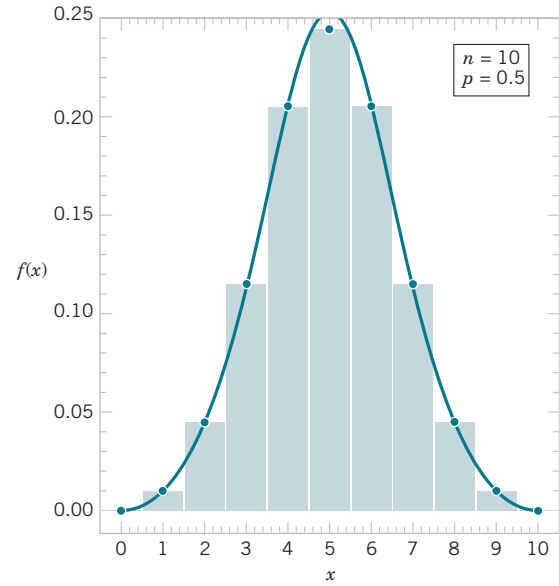
Determining the value of x to meet a specified probability.

4.6

Normal Approximation to the Binomial and Poisson Distributions

We began our section on the normal distribution with the central limit theorem and the normal distribution as an approximation to a random variable with a large number of trials. Consequently, it should not be surprising to learn that the normal distribution can be used to approximate binomial probabilities for cases in which n is large. The following example illustrates that for many physical systems, the binomial model is appropriate with an extremely large value for n . In these cases, it is difficult to calculate probabilities by using the binomial distribution. Fortunately, the **normal approximation** is most effective in these cases. An illustration is provided in Figure 4.16. The area of each bar equals the binomial probability of x . Notice that the area of bars can be approximated by areas under the normal probability density function.

From Figure 4.16, it can be seen that a probability such as $P(3 \leq X \leq 7)$ is better approximated by the area under the normal curve from 2.5 to 7.5. Consequently, a modified interval is used to better compensate for the difference between the continuous normal distribution and the discrete binomial distribution. This modification is called a **continuity correction**.

**FIGURE 4.16**

Normal approximation to the binomial distribution.

EXAMPLE 4.13

Assume that in a digital communication channel, the number of bits received in error can be modeled by a binomial random variable, and assume that the probability that a bit is received in error is 1×10^{-5} . If 16 million bits are transmitted, what is the probability that 150 or fewer errors occur?

Let the random variable X denote the number of errors. Then X is a binomial random variable and

$$P(X \leq 150) = \sum_{x=0}^{150} \binom{16,000,000}{x} (10^{-5})^x (1 - 10^{-5})^{16,000,000-x}$$

Practical Interpretation: Clearly, this probability is difficult to compute. Fortunately, the normal distribution can be used to provide an excellent approximation in this example.

Normal Approximation to the Binomial Distribution

If X is a binomial random variable with parameters n and p ,

$$Z = \frac{X - np}{\sqrt{np(1-p)}} \quad (4.12)$$

is approximately a standard normal random variable. To approximate a binomial probability with a normal distribution, a **continuity correction** is applied as follows:

$$P(X \leq x) = P(X \leq x + 0.5) \approx P\left(Z \leq \frac{x + 0.5 - np}{\sqrt{np(1-p)}}\right)$$

and

$$P(x \leq X) = P(x - 0.5 \leq X) \approx P\left(\frac{x - 0.5 - np}{\sqrt{np(1-p)}} \leq Z\right)$$

The approximation is good for $np > 5$ and $n(1-p) > 5$.

Recall that for a binomial variable X , $E(X) = np$ and $V(X) = np(1 - p)$. Consequently, the expression in Equation 4.12 is nothing more than the formula for standardizing the random variable X . Probabilities involving X can be approximated by using a standard normal distribution. The approximation is good when n is large relative to p .

A way to remember the continuity correction is to write the probability in terms of $\leq$ or $\geq$ and then add or subtract the 0.5 correction factor to make the probability greater.

EXAMPLE 4.14

The digital communication problem in Example 4.13 is solved as follows:

$$\begin{aligned} P(X \leq 150) &= P(X \leq 150.5) \\ &= P\left(\frac{X - 160}{\sqrt{160(1 - 10^{-5})}} \leq \frac{150.5 - 160}{\sqrt{160(1 - 10^{-5})}}\right) \\ &\approx P(Z \leq -0.75) = 0.227 \end{aligned}$$

Because $np = (16 \times 10^6)(1 \times 10^{-5}) = 160$ and $n(1 - p)$ is much larger, the approximation is expected to work well in this case.

Practical Interpretation: Binomial probabilities that are difficult to compute exactly can be approximated with easy-to-compute probabilities based on the normal distribution.

EXAMPLE 4.15 | Normal Approximation to Binomial

Again consider the transmission of bits in Example 4.14. To judge how well the normal approximation works, assume that only $n = 50$ bits are to be transmitted and that the probability of an error is $p = 0.1$. The exact probability that two or fewer errors occur is

$$\begin{aligned} P(X \leq 2) &= \binom{50}{0} 0.9^{50} + \binom{50}{1} 0.1(0.9^{49}) \\ &\quad + \binom{50}{2} 0.1^2(0.9^{48}) = 0.112 \end{aligned}$$

Based on the normal approximation,

$$\begin{aligned} P(X \leq 2) &= P\left(\frac{X - 5}{\sqrt{50(0.1)(0.9)}} \leq \frac{2.5 - 5}{\sqrt{50(0.1)(0.9)}}\right) \\ &\approx P(Z < -1.18) = 0.119 \end{aligned}$$

We can also approximate $P(X = 5)$ as

$$\begin{aligned} P(5 \leq X \leq 5) &= P(4.5 \leq X \leq 5.5) \\ &\approx P\left(\frac{4.5 - 5}{2.12} \leq Z \leq \frac{5.5 - 5}{2.12}\right) \\ &= P(-0.24 \leq Z \leq 0.24) = 0.19 \end{aligned}$$

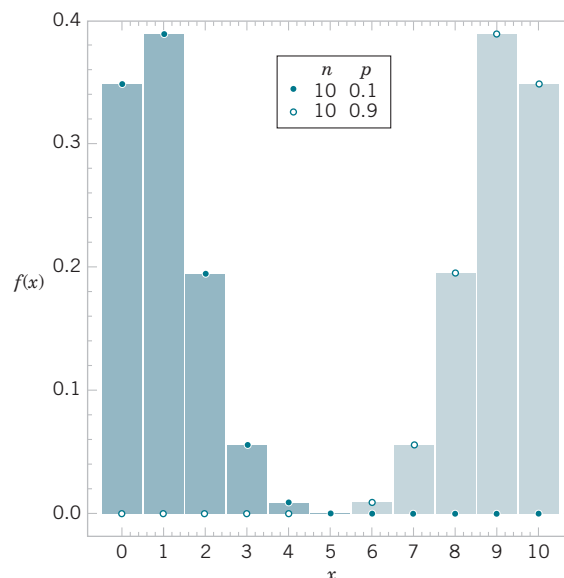
and this compares well with the exact answer of 0.1849.

Practical Interpretation: Even for a sample as small as 50 bits, the normal approximation is reasonable, when $p = 0.1$.

The correction factor is used to improve the approximation. However, if np or $n(1 - p)$ is small, the binomial distribution is quite skewed and the symmetric normal distribution is not a good approximation. Two cases are illustrated in Figure 4.17.

Recall that the binomial distribution is a satisfactory approximation to the hypergeometric distribution when n , the sample size, is small relative to N , the size of the population from which the sample is selected. A rule of thumb is that the binomial approximation is effective if $n/N < 0.1$. Recall that for a hypergeometric distribution, p is defined as K/N . That is, p is interpreted as the number of successes in the population. Therefore, the normal distribution can provide an effective approximation of hypergeometric probabilities when $n/N < 0.1$, $np > 5$, and $n(1 - p) > 5$. Figure 4.18 provides a summary of these guidelines.

Recall that the Poisson distribution was developed as the limit of a binomial distribution as the number of trials increased to infinity. Consequently, it should not be surprising to find that the normal distribution can also be used to approximate probabilities of a Poisson random variable.

**FIGURE 4.17**

Binomial distribution is not symmetrical if p is near 0 or 1.

Normal Approximation to the Poisson Distribution

If X is a Poisson random variable with $E(X) = \lambda$ and $V(X) = \lambda$,

$$Z = \frac{X - \lambda}{\sqrt{\lambda}} \quad (4.13)$$

is approximately a standard normal random variable. The same continuity correction used for the binomial distribution can also be applied. The approximation is good for

$$\lambda > 5$$

EXAMPLE 4.16 | Normal Approximation to Poisson

Assume that the number of asbestos particles in a squared meter of dust on a surface follows a Poisson distribution with a mean of 1000. If a squared meter of dust is analyzed, what is the probability that 950 or fewer particles are found?

This probability can be expressed exactly as

$$P(X \leq 950) = \sum_{x=0}^{950} \frac{e^{-1000} 1000^x}{x!}$$

The computational difficulty is clear. The probability can be approximated as

$$\begin{aligned} P(X \leq 950) &= P(X \leq 950.5) \approx P\left(Z \leq \frac{950.5 - 1000}{\sqrt{1000}}\right) \\ &= P(Z \leq -1.57) = 0.058 \end{aligned}$$

Practical Interpretation: Poisson probabilities that are difficult to compute exactly can be approximated with easy-to-compute probabilities based on the normal distribution.

Hypergeometric distribution	$\frac{n}{N} \approx < 0.1$	Binomial distribution	$np > 5$ $n(1-p) > 5$	Normal distribution
--------------------------------	-----------------------------	--------------------------	--------------------------	------------------------

FIGURE 4.18

Conditions for approximating hypergeometric and binomial probabilities.

4.7 Exponential Distribution

The discussion of the Poisson distribution defined a random variable to be the number of flaws along a length of copper wire. The distance between flaws is another random variable that is often of interest. Let the random variable X denote the length from any starting point on the wire until a flaw is detected. As you might expect, the distribution of X can be obtained from knowledge of the distribution of the number of flaws. The key to the relationship is the following concept. The distance to the first flaw exceeds 3 millimeters if and only if there are no flaws within a length of 3 millimeters—simple but sufficient for an analysis of the distribution of X .

In general, let the random variable N denote the number of flaws in x millimeters of wire. If the mean number of flaws is λ per millimeter, N has a Poisson distribution with mean λx . We assume that the wire is longer than the value of x . Now,

$$P(X > x) = P(N = 0) = \frac{e^{-\lambda x}(\lambda x)^0}{0!} = e^{-\lambda x}$$

Therefore,

$$F(x) = P(X \leq x) = 1 - e^{-\lambda x}, \quad x \geq 0$$

is the cumulative distribution function of X . By differentiating $F(x)$, the probability density function of X is calculated to be

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

The derivation of the distribution of X depends only on the assumption that the flaws in the wire follow a **Poisson process**. Also, the starting point for measuring X does not matter because the probability of the number of flaws in an interval of a Poisson process depends only on the length of the interval, not on the location. For any Poisson process, the following general result applies.

Exponential Distribution

The random variable X that equals the distance between successive events from a Poisson process with mean number of events $\lambda > 0$ per unit interval is an **exponential random variable** with parameter λ . The probability density function of X is

$$f(x) = \lambda e^{-\lambda x} \quad \text{for} \quad 0 \leq x < \infty \quad (4.14)$$

The exponential distribution obtains its name from the exponential function in the probability density function. See plots of the exponential distribution for selected values of λ in Figure 4.19. For any value of λ , the exponential distribution is quite skewed. The following results are easily obtained and are left as an exercise.

Mean and Variance

If the random variable X has an exponential distribution with parameter λ ,

$$\mu = E(X) = \frac{1}{\lambda} \quad \text{and} \quad \sigma^2 = V(X) = \frac{1}{\lambda^2} \quad (4.15)$$

It is important to *use consistent units* to express intervals, X , and λ . The following example illustrates unit conversions.

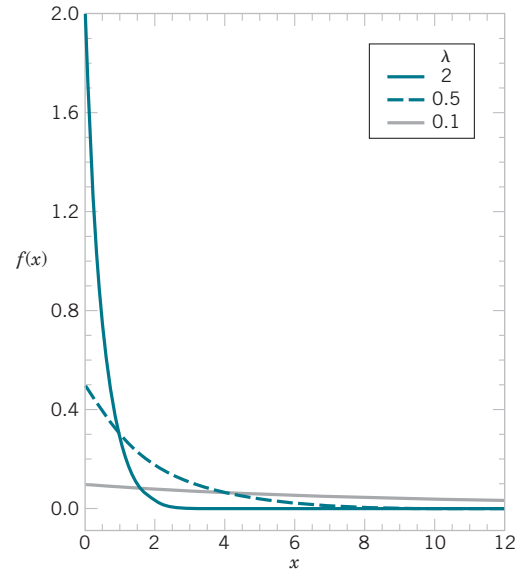


FIGURE 4.19

Exponential probability density functions for selected values of λ .

EXAMPLE 4.17 | Computer Usage

In a large corporate computer network, user log-ons to the system can be modeled as a Poisson process with a mean of 25 log-ons per hour. What is the probability that there are no log-ons in an interval of 6 minutes?

Let X denote the time in hours from the start of the interval until the first log-on. Then X has an exponential distribution with $\lambda = 25$ log-ons per hour. We are interested in the probability that X exceeds 6 minutes. Because λ is given in log-ons per hour, we express all time units in hours. That is, 6 minutes = 0.1 hour. Therefore,

$$P(X > 0.1) = \int_{0.1}^{\infty} 25e^{-25x} dx = e^{-25(0.1)} = 0.082$$

The cumulative distribution function can also be used to obtain the same result as follows:

$$P(X > 0.1) = 1 - F(0.1) = e^{-25(0.1)}$$

An identical answer is obtained by expressing the mean number of log-ons as 0.417 log-ons per minute and computing the probability that the time until the next log-on exceeds 6 minutes. Try it.

What is the probability that the time until the next log-on is between 2 and 3 minutes? Upon converting all units to hours,

$$\begin{aligned} P(0.033 < X < 0.05) &= \int_{0.033}^{0.05} 25e^{-25x} dx \\ &= -e^{-25x} \Big|_{0.033}^{0.05} = 0.152 \end{aligned}$$

An alternative solution is

$$P(0.033 < X < 0.05) = F(0.05) - F(0.033) = 0.152$$

Determine the interval of time such that the probability that no log-on occurs in the interval is 0.90. The question asks for the length of time x such that $P(X > x) = 0.90$. Now,

$$P(X > x) = e^{-25x} = 0.90$$

Take the (natural) log of both sides to obtain $-25x = \ln(0.90) = -0.1054$. Therefore,

$$x = 0.00421 \text{ hour} = 0.25 \text{ minute}$$

Furthermore, the mean time until the next log-on is

$$\mu = 1/25 = 0.04 \text{ hour} = 2.4 \text{ minutes}$$

The standard deviation of the time until the next log-on is

$$\sigma = 1/25 \text{ hours} = 2.4 \text{ minutes}$$

Practical Interpretation: Organizations make wide use of probabilities for exponential random variables to evaluate resources and staffing levels to meet customer service needs.

In Example 4.17, the probability that there are no log-ons in a six-minute interval is 0.082 regardless of the starting time of the interval. A Poisson process assumes that events occur uniformly throughout the interval of observation; that is, there is no clustering of events. If the log-ons are well modeled by a Poisson process, the probability that the first log-on after noon occurs after 12:06 P.M. is the same as the probability that the first log-on after 3:00 P.M. occurs after 3:06 P.M. And if someone logs on at 2:22 P.M., the probability that the next log-on occurs after 2:28 P.M. is still 0.082.

Our starting point for observing the system does not matter. However, if high-use periods occur during the day, such as right after 8:00 A.M., followed by a period of low use, a Poisson process is not an appropriate model for log-ons and the distribution is not appropriate for computing probabilities. It might be reasonable to model each of the high- and low-use periods by a separate Poisson process, employing a larger value for λ during the high-use periods and a smaller value otherwise. Then an exponential distribution with the corresponding value of λ can be used to calculate log-on probabilities for the high- and low-use periods.

Lack of Memory Property An even more interesting property of an exponential random variable concerns conditional probabilities.

EXAMPLE 4.18 | Lack of Memory Property

Let X denote the time between detections of a particle with a Geiger counter and assume that X has an exponential distribution with $E(X) = 1.4$ minutes. The probability that we detect a particle within 0.5 minute of starting the counter is

$$P(X < 0.5) = F(0.5) = 1 - e^{-0.5/1.4} = 0.30$$

Now, suppose that we turn on the Geiger counter and wait 3 minutes without detecting a particle. What is the probability that a particle is detected in the next 30 seconds?

Because we have already been waiting for 3 minutes, we feel that a detection is “due.” That is, the probability of a detection in the next 30 seconds should be higher than 0.3. However, for an exponential distribution, this is not true. The requested probability can be expressed as the conditional probability that $P(X < 3.5 | X > 3)$. From the definition of conditional probability,

$$P(X < 3.5 | X > 3) = P(3 < X < 3.5) / P(X > 3)$$

where

$$\begin{aligned} P(3 < X < 3.5) &= F(3.5) - F(3) \\ &= [1 - e^{-3.5/1.4}] - [1 - e^{-3/1.4}] = 0.035 \end{aligned}$$

and

$$P(X > 3) = 1 - F(3) = e^{-3/1.4} = 0.117$$

Therefore,

$$P(X < 3.5 | X > 3) = 0.035 / 0.117 = 0.30$$

Practical Interpretation: After waiting for 3 minutes without a detection, the probability of a detection in the next 30 seconds is the same as the probability of a detection in the 30 seconds immediately after starting the counter. The fact that we have waited 3 minutes without a detection does not change the probability of a detection in the next 30 seconds.

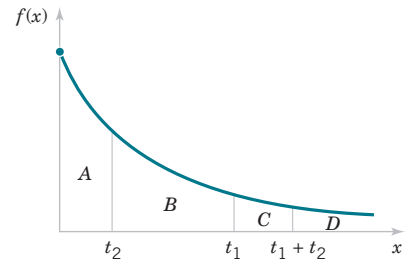
Example 4.18 illustrates the **lack of memory property** of an exponential random variable, and a general statement of the property follows. In fact, the exponential distribution is the only continuous distribution with this property.

Lack of Memory Property

For an exponential random variable X ,

$$P(X < t_1 + t_2 | X > t_1) = P(X < t_2) \quad (4.16)$$

Figure 4.20 graphically illustrates the lack of memory property. The area of region A divided by the total area under the probability density function ($A + B + C + D = 1$) equals $P(X < t_2)$.

**FIGURE 4.20****Lack of memory property of an exponential distribution.**

The area of region C divided by the area $C + D$ equals $P(X < t_1 + t_2 \mid X > t_1)$. The lack of memory property implies that the proportion of the total area that is in A equals the proportion of the area in C and D that is in C .

The lack of memory property is not so surprising when we consider the development of a Poisson process. In that development, we assumed that an interval could be partitioned into small intervals that were independent. These subintervals are similar to independent Bernoulli trials that comprise a binomial experiment; knowledge of previous results does not affect the probabilities of events in future subintervals. An exponential random variable is the continuous analog of a geometric random variable, and it shares a similar lack of memory property.

The exponential distribution is often used in reliability studies as the model for the time until failure of a device. For example, the lifetime of a semiconductor chip might be modeled as an exponential random variable with a mean of 40,000 hours. The lack of memory property of the exponential distribution implies that the device does not wear out. That is, regardless of how long the device has been operating, the probability of a failure in the next 1000 hours is the same as the probability of a failure in the first 1000 hours of operation. The lifetime L of a device with failures caused by random shocks might be appropriately modeled as an exponential random variable.

However, the lifetime L of a device that suffers slow mechanical wear, such as bearing wear, is better modeled by a distribution such that $P(L < t + \Delta t \mid L > t)$ increases with t . Distributions such as the Weibull distribution are often used in practice to model the failure time of this type of device. The Weibull distribution is presented in a later section.

4.8**Erlang and Gamma Distributions**

An exponential random variable describes the length until the first event is obtained in a Poisson process. A generalization of the exponential distribution is the length until r events occur in a Poisson process.

EXAMPLE 4.19 | Processor Failure

The failures of the central processor units of large computer systems are often modeled as a Poisson process. Typically, failures are not caused by components wearing out but by more random failures of the large number of semiconductor circuits in the units. Assume that the units that fail are immediately repaired and assume that the mean number of failures per hour is 0.0001. Let X denote the time until four failures occur in a system. Determine the probability that X exceeds 40,000 hours.

Let the random variable N denote the number of failures in 40,000 hours of operation. The time until four failures occur exceeds 40,000 hours if and only if the number of failures in 40,000 hours is three or less. Therefore,

$$P(X > 40,000) = P(N \leq 3)$$

The assumption that the failures follow a Poisson process implies that N has a Poisson distribution with

$$E(N) = \lambda T = 0.0001(40,000) = 4$$

failures per 40,000 hours. Therefore,

$$P(X > 40,000) = P(N \leq 3) = \sum_{k=0}^3 \frac{e^{-4} 4^k}{k!} = 0.433$$